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The Effect of Benzodiazepine Hypnotics and their Metabolites on Somatic Motor Control and Proprioceptive Function in Elderly Patients

Abstract:

It has been estimated that over 20 billion dollars is spent annually on falls and fall related injury in the United States alone³. Falls are most common among the elderly, and it is estimated that 20-30% of falls in elderly patients incur some sort of fracture or other trauma³. In addition to this, a large percentage of elderly citizens in the United States are being prescribed benzodiazepines and other hypnotics as sleep aids. Literature states that depression and other common disorders of the elderly are frequently misdiagnosed and subsequently mistreated with benzodiazepines. In conjunction with this, although health experts recommend that hypnotics should not be prescribed for more than 2-3 weeks, benzodiazepines are typically taken for months or even years at a time¹. In this research article, the use of benzodiazepine hypnotics is clearly implicated as a cause for proprioceptive dysfunction and postural instability. To prove that there is a clear effect of benzodiazepine hypnotics on the proprioceptive abilities of elderly patients, molecular mechanisms of benzodiazepine use on proprioceptive function are explored. Because the many metabolites of benzodiazepines are considered pharmacologically active, the effects of these solitary metabolites on the proprioceptive function of patients are explored. Finally, alternatives to benzodiazepine use are investigated as well as suggestions for future study.

Introduction:

The incidence of falls and fall related injury in elderly patients has reached epidemic proportions in the United States and around the world. In addition to this, over two thirds of elderly patients that were admitted to the hospital admitted to chronic or intermittent use of benzodiazepines or other hypnotics within the past year. With these things in mind, in culmination with the fact that most of the elderly are considered to be under the poverty level and therefore paying for these injuries with Medicaid or similar governmental programs, a severe look has to be taken into the continued prescription practices of the physicians for these hypnotic drugs.

In order to illicit a strong biological basis for how psychoactive drug use can inhibit proprioceptive function and subsequent motor control, samples of elderly patients were assessed in varying tests of balance and motor ability. Statistical analysis was performed on the results, and it was found that psychoactive hypnotics were implicated with a loss of postural stability and proprioceptive function in these patients.

Due to the long half-life and high pharmacological activity of the metabolites of benzodiazepines *in vivo*, the effect of these various *metabolites* on proprioceptive function needs to be elucidated.

Therefore, the effects of these various hypnotic metabolites on proprioceptive function were investigated in experiments where balance and postural stability were measured subsequent to metabolite administration. The effects of these compounds alone and in combination with alcohol were measured for significant effect on motor function against placebo groups and the parent compound. It was found that while there is a significant buildup of these metabolically active compounds over time, none of the metabolites of benzodiazepines have a significant impact on balance or proprioceptive function. The loss of postural stability in human subjects is therefore due to the function of the parent compound on these subjects.

Because molecular biological studies and gene knockout experiments on human subjects are “frowned upon” studies elucidating the specific molecular mechanism of benzodiazepine hypnotics and human proprioceptive function could not be found. Therefore, the molecular mechanisms for benzodiazepine drugs on proprioceptive function were researched in supplement and a mechanism was proposed for the action of these compounds on proprioceptors and subsequent somatic motor function. This information is provided in the discussion section.

Finally, suggestions for physicians on how to curb the prescription rates of benzodiazepines are explored along with useful strategies for how to recognize dependence on the hypnotic compounds, and alternatives to the prescription of these compounds for insomnia and depression related conditions are suggested.

Methods/Results:

The first test that is needed in establishing if there is a relationship between these hypnotics and a loss of postural stability is the direct measurement of the administration of the parent drugs themselves and the loss of proprioception and somatic motor function. In this the authors hypothesized that administration of benzodiazepines would in fact have an effect on somatic motor control and proprioceptive ability. The control group for this study was a total of 338 women that were subjected to these tests that were not taking any psychoactive medications. In addition 64 women taking one psychoactive medication and 12 women taking two psychoactive medications were tested. The mean average for the age of the participants was in the mid-seventies for each of the groups tested. The standard deviation for each of these groups was around 6, too. This is good because it means that random youngsters were not being thrown into groups and subsequently skewing the results towards a false statistical significance.

In order to prove the correlation between benzodiazepine hypnotics and a loss of postural stability, two tests of somatic motor function were administered to the subjects subsequent to drug or placebo administration. Body sway was measured using a sway meter that measured the displacement of the body in 30 second intervals. There were four conditions for this test; body sway was measured on a firm surface with subject's eyes being open, body sway measured on the same apparatus with the subject's eyes being closed, and then on a soft surface (foam pad) with the subject's eyes again being open and closed. The amount of sway measured in millimeters was measured and then averaged for each trial run. Next the dynamic balance of the subjects was measured by making the subjects walk in place for 1 minute with their eyes closed. This test was again assessed on a scale of one to five. A composite score was then taken by calculating the mean average of the scores for each test, and statistical analysis was taken to determine if the differences between the nil medications, the one medication, and two medication

groups were statistically significant. The two different tests on somatic motor function and proprioceptive ability are significant because both static standing balance and dynamic walking balance are significant in daily life. The statistical test run on these experiments was a two way analysis of variance. This test was used because there are multiple causative factors that are affecting the postural stability of the subjects (the differing age of individual subjects and the amount/type of psychoactive medications).

The results of these tests were shown in Figure 1. All numbers are statistically significant; meaning the p-value calculated for each trial was under .05.

Figure 1:

	<i>Nil (n = 338)</i> <i>Mean (s.d.)</i>	<i>One (n = 64)</i> <i>Mean (s.d.)</i>	<i>Two (n = 12)</i> <i>Mean (s.d.)</i>	<i>Total (n = 414)</i> <i>Mean (s.d.)</i>
<i>Sway^f</i>				
Eyes open (floor)	81 (43)	88 (48)	128 (141)	83 (49)
Eyes closed (floor)	115 (61)	117 (67)	123 (57)	116 (62)
Eyes open (foam)	132 (65)	147 (74)	169 (71)	136 (67)
Eyes closed (foam)	234 (109)	237 (127)	262 (103)	235 (112)

Taken from Lord et.al. 1995.

Measurements of sway were measured in millimeters in 30 second intervals. N is representative of the number of subjects used for each individual trial. (N=338 would correspond to 338 people being part of that trial.)

From figure one, one can tell that the effect of benzodiazepines on sway, though minimal for the eyes open floor group were extremely pronounced for the eyes open foam group. In each case, differences in sway were more pronounced in the two group than they were in the one group. The “Two” group corresponds to people who took two benzodiazepine hypnotics at the time of the trial instead of one. Something interesting to note is that although there is typically a large difference between the eyes open groups and the eyes closed groups, there is not typically a large difference between the nil, one and two groups *within* the eyes closed. In other words, although the subjects did much worse at standing with their eyes closed than they did standing with their eyes open, this was not significantly worsened by them being on hypnotic medications.

The assessment of benzodiazepine metabolites on proprioceptive function is important because it has been shown that many of these metabolites are pharmacologically active and have long half-lives. This in conjunction with the fact that the current lackadaisical prescription protocol for benzodiazepines (they are often over prescribed or prescribed way longer than what is actually recommended for healthcare professionals) could mean that there is a buildup of these pharmacologically active compounds in patients’ bodies over time. Therefore, if there is a deleterious effect of these metabolites on somatic motor control and proprioceptive function, than patients could be affected hours or even days after the parent compound has worn off. In elderly patients who use these compounds as sleep aids, this could mean that balance and stability could be affected if they get up to go to the bathroom in the middle of the night, or even the next day. The authors’ hypotheses were that these metabolites would in fact have an effect on proprioceptive function, especially in conjunction with alcohol. An important note for this study is that the authors tested healthy young volunteers. While this is significant in determining if the metabolites of benzodiazepines have an effect on the balance and postural stability, the effect of these metabolites on the senior population should also be investigated. This is because there is a

slowdown and loss of postural stability related with senescence already so the effect of these compounds could be proportionally or even exponentially larger when tested in the elderly population.

To test the effect of the *metabolites* of benzodiazepines on proprioceptive function and psychomotor activity, 40 healthy student volunteers aged 20-29 were divided into two groups. One group was instructed to take oxazepam and methyloxazepam at 15 and 20 mg respectively for two weeks. The other group received N-desmethyldiazepam and chlordiazepoxide lactum and placebo for two weeks at a dosage of 5mg and 10 mg respectively. These compounds are the main metabolites of the two drugs diazepam and chlórdiazepoxide. Blood samples were taken in the 1st, 7th, and 14th days of administration of these compounds to measure the concentration in the blood plasma of the subjects. Driving coordination, proprioception, and attention were also measured on the 1st, 7th, and 14th days following administration of the compounds. Statistical analysis was performed on the results and it was found that the results were indeed significant; however, there was a minimal effect of any of these metabolites on the somatic control of motor movements and proprioception.

Testing the statistical significance of such a study is crucial because we cannot do knock out studies in human subjects, or control for factors non related to the study. For example, in a laboratory setting, the mice used for a specific test are all the exact same age and grow up in the exact same conditions. In these studies; however, the human subjects come from all backgrounds, and differ in age by as much as a decade. Therefore, statistical analysis is not only necessary, but also makes up the bulk of our correlational study.

Chlórdiazepoxide lactum was the only drug that had impaired co-ordination and increased the number of mistakes in the trials without alcohol, and even this increase was minimal. The only results that were found to be different than the placebo groups for these variables were when the various metabolites were combined with alcohol, however, none of these showed a difference larger than the alcohol itself. In other words, no correlation between the concentration of the metabolites in the patients' blood and an impairment of skills was evident. The results of this study are shown on Figure 2. There was also no increase with the deleterious of these drugs as they built up over time. In fact, there was an improvement in each of the motor tasks as time went on, despite the metabolites still being in the patients' systems. This meant that although the metabolites were accumulating in the subject's bodies over time (Figure 3), there was not obvious effect on proprioception or proprioceptive ability. Furthermore, subjects who were

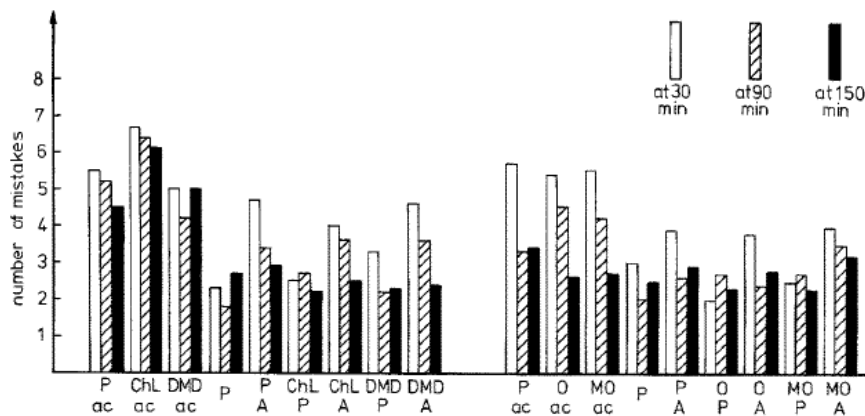


Fig. 1. Number of mistakes in co-ordination test I. According to two way analysis of variance the effect of treatment was significant at the $p < 0.001$ level

P: Placebo
ChL: chlordiazepoxide lactam
DMD: N-desmethyldiazepam
O: Oxazepam
MO: Methyloxazepam
ac: Placebo drink
A: Alcohol
for further explanation: see text.

Figure 2: Taken from Palva et. al. 1978.

administered chlorodiazepoxide lactum, went back down to placebo levels after 150 minutes of being administered the drug and their error rate stayed at placebo levels after that which suggests that although there was an initial effect, the subjects of the experiment quickly developed a tolerance for the metabolite and were able to return to normal function. Table 1 provides a summary of the metabolites that accumulated in the patients' systems over time in nanograms/ml. Although this information is quite disconcerting because these metabolites are known to be pharmacologically active, there was not apparent effect on proprioception or somatic motor control.

Table 1. Serum levels of the drugs on the 1st, 7th and 14th days of treatment, three hours after administration of the last capsule. For symbols see text

Drug	Serum level of drug ng/ml (mean $\pm$ SD)		
	1st day	7th day	14th day
ChL	430 $\pm$ 50	2550 $\pm$ 1220	2700 $\pm$ 1500
DMD	82 $\pm$ 24	920 $\pm$ 440	1220 $\pm$ 650
O	91 $\pm$ 73	270 $\pm$ 146	219 $\pm$ 111
MO	270 $\pm$ 120	710 $\pm$ 350	770 $\pm$ 490

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Table 1: Taken from: Palva et. al. 1978.

Discussion:

The results from Lord et. al. suggest that more care has to be taken in prescribing benzodiazepine hypnotics to elderly patients. Sway tests indicated that elderly patients perform significantly worse after taking benzodiazepines. We know from literature that proprioceptive function works off of mechano-stretch receptors in the spinal cord. As muscles stretch, muscle

spindles oriented in parallel with muscle fibers and as the muscle is stretched, the firing rate of these muscle fibers goes up. Although there is a lack of molecular biological studies done in human subjects to illicit the exact function of hypnotic compounds on proprioceptive function, it is known that Benzodiazepines are known GABA_A agonists.³ Hypnotic compounds therefore must work on GABA_A receptors in the spinal cord to hyperpolarize I_a neurons and inhibit the function of these reflexes. It was also shown in Palva et. al. that the active metabolites of benzodiazepine hypnotics do not exhibit the same inhibition on proprioceptive function that their parent compounds exhibit. This must mean that these compounds although shown to be pharmacologically active in other processes⁴ no longer bind to these GABA_A receptors and cause the hyperpolarization of I_a neurons. In the interest of further experimentation, the effect of these metabolites on the somatic motor control should also be studied in elderly patients because of the known effects of senescence on both balance and rates of metabolism.¹

Given the known effects of benzodiazepine hypnotics on proprioception in elderly patients, it is therefore suggested that physicians curtail their scripting of these drugs. Suggestions for this, detailed in Huston, 1992 include more rigorous tests for depression and other conditions before writing prescriptions for benzodiazepenes, and suggesting alternatives for benign sleep disturbances. If the patient is persistent, the physician should suspect physician shopping and if necessary refer the patient to a likeminded colleague¹

For further examination on the effects of these drugs on proprioceptive systems, I would create a mouse model knock out for GABA_A receptors in the spinal cord in mice via Cre-Lox cellular machinery. Then I would test to see if benzodiazepine hypnotics had the same function in these mouse models.

References:

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